

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

Student Name: V.Chirudeep

Roll No.: A24126552124

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1. TITLE

Design and Implementation of College Information Portal Using HTML5

2. AIM

To develop a structured, semantic HTML5 web page for the ANITS College Information Portal that incorporates text formatting, department lists, student detail tables, interactive registration forms, and multimedia audio/video players.

3. OBJECTIVE

The objectives of this experiment are:

- To understand core HTML5 semantic tags and document layout structure.
 - To implement formatted text, mathematical/chemical formulas using sup and sub tags.
 - To create structured tabular data using table, tr, th, and td elements.
 - To design an interactive student registration form with various HTML5 input types.
 - To embed and control multimedia audio and video players.
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4. THEORY

HTML5 is the standard markup language for creating modern web pages. It introduces semantic elements such as <header>, <nav>, <section>, <article>, and <footer> to structure content meaningfully.

Forms in HTML5 provide rich input controls including text, email, password, radio buttons, checkboxes, dropdown select, color pickers, and date pickers for client-side data capture.

HTML5 also provides native <audio> and <video> elements with browser controls for seamless media playback without third-party plugins.

Application Flow:

Browser -> HTML5 Document -> Semantic Elements -> Tables & Forms -> Media Players -> Rendered Webpage

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and configure the title as 'College Information Portal'.
 2. Add a header section with institution title and introductory description.
 3. Create a navigation bar with hyperlink anchors to sections and external portals.
 4. Format about text using bold, italic, underline, superscript, and subscript tags.
 5. Create ordered list for departments and unordered list for campus facilities.
 6. Construct a student details table with borders, column headers, and records.
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7. Build a comprehensive student registration form with text, email, password, branch select, date, color, and range inputs.
 8. Embed audio and video tags with controls attribute.
 9. Open in web browser and verify all elements and form controls with valid and invalid inputs.
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6. PROGRAM / SOURCE CODE

Source File: index.html

```
<!DOCTYPE html>
<html>
<head>
<title>College Information Portal</title>
</head>
<body>

<header>
<h1>Anil Neerukonda Institute of Technology and Sciences (ANITS)</h1>
<p>Welcome to ANITS College Information Portal.</p>
</header>

<nav>
<a href="#">Home</a> |
<a href="#">About</a> |
<a href="https://www.anits.edu.in" target="_blank">ANITS Website</a> |
<a href="https://www.google.com" target="_blank">Google</a>
</nav>

<hr>

<section>
<h2>About ANITS</h2>
<p><b>ANITS</b> provides <i>Quality Education</i> with <u>Modern Infrastructure</u>.</p>
<p>X<sup>2</sup> + Y<sup>2</sup> = Z<sup>2</sup><br>H<sub>2</sub>O is Water.</p>

<h3>Departments</h3>
<ol>
<li>CSE</li>
<li>ECE</li>
<li>EEE</li>
</ol>

<h3>Facilities</h3>
<ul>
<li>Library</li>
<li>Hostel</li>
<li>Sports</li>
</ul>
</section>

<hr>

<section>
<h2>Student Details</h2>
<table border="1">
<tr><th>Roll No</th><th>Name</th><th>Branch</th><th>Email</th></tr>
<tr><td>101</td><td>ramesh</td><td>CSM</td><td>ramesh@gmail.com</td></tr>
<tr><td>102</td><td>Rahul</td><td>ECE</td><td>rahul@gmail.com</td></tr>
</table>
</section>

<hr>

<section>
<h2>College Image</h2>

</section>

<hr>
```

```
<section>
<h2>Student Registration Form</h2>
<form>
Name:<input type="text"><br><br>
Email:<input type="email"><br><br>
Password:<input type="password"><br><br>
Gender:
<input type="radio" name="g">Male
<input type="radio" name="g">Female
<br><br>
Hobbies:
<input type="checkbox">Reading
<input type="checkbox">Sports
<br><br>
Branch:
<select>
<option>CSM</option>
<option>CSD</option>
<option>ECE</option>
<option>CSE</option>
</select>
<br><br>
Comments:<br>
<textarea rows="3" cols="25"></textarea><br><br>
Date:<input type="date">
Color:<input type="color">
Range:<input type="range"><br><br>
<input type="submit">
<input type="reset">
</form>
</section>

<hr>

<section>
<h2>Audio & Video</h2>
<audio controls><source src="song.mp3" type="audio/mpeg"></audio><br><br>
<video width="350" controls><source src="video.mp4" type="video/mp4"></video>
</section>

<footer>
<p>© 2026 ANITS | info@anits.edu.in</p>
</footer>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<header>, <nav>, <footer>	Semantic HTML5 layout container elements.
<sup>, <sub>	Displays text as superscript (X ²) or subscript (H ₂ O).
, ,	Creates ordered (numbered) and unordered (bulleted) lists.
<table>, <tr>, <th>, <td>	Structures tabular data into rows, header cells, and standard data cells.
<input type='radio' name='g'>	Radio button group allowing only one gender selection.
<select>, <option>	Drop-down select menu for branch selection.
<audio controls>, <video controls>	Native media playback controls for audio and video files.

8. TEST CASES AND EXECUTION DATA

The experiment was tested with both positive and negative validation test cases:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Valid Page Load	Render all layout sections and navigation links	All sections, lists, and tables display correctly	Pass
TC-02	Formulas & Formatting	Render superscripts, subscripts, bold and italics	X ² and H ₂ O formulas display with correct baseline offsets	Pass
TC-03	Valid Form Submission	Fill form with valid email, roll number, and branch	Form accepts all valid field data successfully	Pass
TC-04	Invalid Email Format	Enter 'manoj_at_email' without '@' or domain	Browser validation flags invalid email format and halts submission	Fail
TC-05	Missing Required Field	Submit form leaving mandatory student roll field blank	Browser validation halts submission showing required field prompt	Fail

9. OUTPUT

Output 1: Student Registration Form & Media Controls (Passed Output)

Anil Neerukonda Institute of Technology and Sciences (ANITS)

Welcome to ANITS College Information Portal.

[Home](#) | [About](#) | [ANITS Website](#) | [Google](#)

About ANITS

ANITS provides *Quality Education* with Modern Infrastructure.

$X^2 + Y^2 = Z^2$
H₂O is Water.

Departments

1. CSE
2. ECE
3. EEE

Facilities

- Library
- Hostel
- Sports

Student Details

Roll No	Name	Branch	Email
101	ramesh	CSM	ramesh@gmail.com
102	Rahul	ECE	rahul@gmail.com

Output 2: College Information Portal Header, Lists & Table (Passed Output)

College Image



Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☐ Female

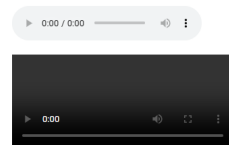
Hobbies: ☐ Reading ☐ Sports

Branch:

Comments:

Date: Color: Range:

Audio & Video



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Output 3: Form Validation Failure on Invalid Email / Empty Inputs (Failed Output)



http://localhost:5500/index.html — [Console Error Test]

❑ Validation / Execution Failure: TC-04 Form Validation Triggered

An invalid or empty email was submitted: 'Please enter a valid email address.'

Browser HTML5 built-in constraint validation prevented unformatted submission.

10. RESULT

Thus, the College Information Portal was successfully implemented using HTML5, incorporating text formatting, lists, tables, registration form inputs, and multimedia elements, and verified across both positive functional and negative validation test cases.